

PASSERINES

This huge order makes up around 60 per cent of all bird species. They are known collectively as perching birds, for their specialized feet.

Like many birds, passerines have four-toed feet with three toes pointing forwards and one back. As a passerine lands, a muscle automatically tightens tendons running through the legs so that the toes lock around a perch, securing the bird even in sleep.

These birds, of the order Passeriformes, are found in almost all land habitats worldwide, from dense rainforest to arid desert and even freezing Arctic tundra. They range in size from the tiny American flycatchers, no bigger than hummingbirds, to the powerfully built Eurasian ravens.

DIVERSITY

The order includes birds adapted to a variety of feeding habits. The insect-eaters have a bill like a needle for probing foliage or with a wide gape for catching prey in flight. Other passerines have a stubby bill for cracking open seeds, while some have a long curved bill for extracting nectar.

Passerines combine a high metabolic rate with a comparatively large brain size, giving some the resilience to tolerate cold and a few the intelligence to master the use of simple tools. They produce helpless, naked young, which are reared in nests ranging from simple cups to elaborate mud-chambers or hanging purses of woven grass. A few passerines are brood parasites, laying their eggs in the nests of other species.

SONGBIRDS

Passerines are split into two groups, largely on the basis of the structure of their vocal box. The first group, the Suboscines, forms around a fifth of all passerines. It occurs in the Old World tropics but has greatest diversity in America; it includes broadbills, pittas, antbirds, and tyrant flycatchers. The second group, the Oscines, includes all remaining passerine families, popularly known as songbirds. Birds often communicate by voice, but the structure of their vocal box enables many passerines to deliver complex songs that are important in courtship and defending territory.

Songs are often so distinctive that they can be used to identify species.

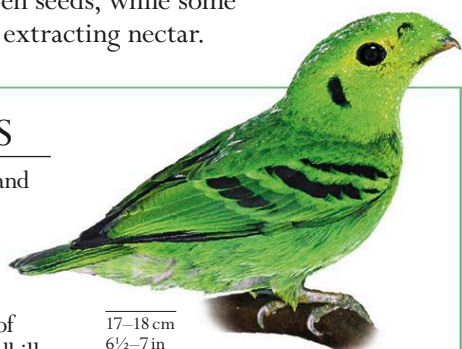
PHYLUM	CHORDATA
CLASS	AVES
ORDER	PASSERIFORMES
FAMILIES	141
SPECIES	6,456

DEBATE OUT OF AUSTRALIA

Early DNA analysis identified two songbird groups: Corvida and Passerida. The mostly Old World Corvida includes carnivores and fruit-eaters, such as crows and birds of paradise. The global Passerida – with more than a quarter of all bird species – includes tits, sunbirds, and finches. Later techniques revealed several primitive Australian groups, such as the lyrebirds, which do not belong to either of these divisions. Together with fossil evidence, this suggests that songbirds diversified in Australasia before giving rise to Corvida and Passerida.

BROADBILLS

The families Calyptomenidae and Eurylaimidae consist of forest-dwelling birds from tropical Africa and Asia. They mostly use their wide bill to catch insects in trees, but members of one group of Asian green broadbills are fruit-eaters.



17–18 cm
6½–7 in

GREEN BROADBILL

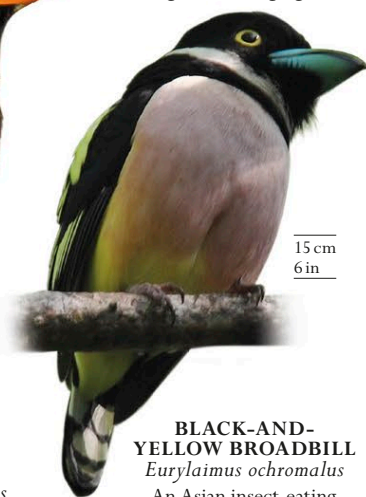
Calyptomena viridis
One of the three S.E. Asian green broadbills, this bird is a specialist fruit-eater and builds globular hanging nests.



25 cm
10 in

BLACK-AND-RED BROADBILL

Cymbirhynchus macrorhynchos
This distinctively coloured S.E. Asian broadbill frequents forests near water. It builds a pouch-like nest suspended from the tips of branches.



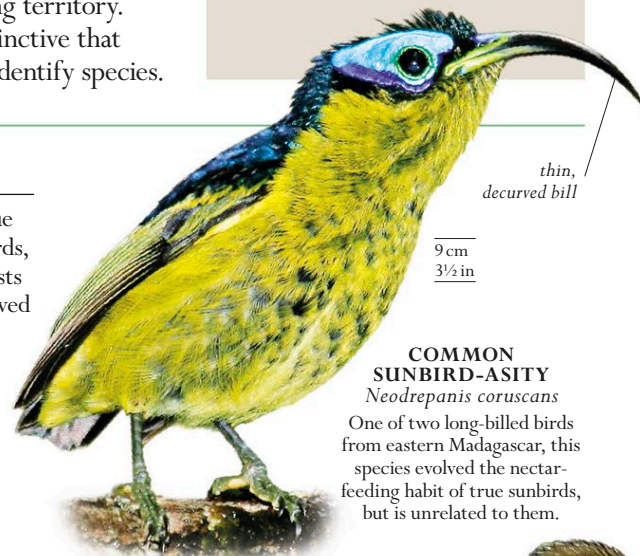
15 cm
6 in

BLACK-AND-YELLOW BROADBILL

Eurylaimus ochromalus
An Asian insect-eating broadbill, this species forages in the middle and upper levels of rainforests from Myanmar to Borneo and Sumatra.

ASITIES

The brush-tipped tongue of these Madagascan birds, the Philepittidae, suggests that they may have evolved from nectar-feeding ancestors: one genus is now fruit-eating; the other resembles the unrelated sunbirds that feed on nectar.



9 cm
3½ in

COMMON SUNBIRD-ASITY

Neodrepanis coruscans
One of two long-billed birds from eastern Madagascar, this species evolved the nectar-feeding habit of true sunbirds, but is unrelated to them.

PITTAS

Birds of this Old World family forage for insects on the forest floor in tropical regions. The Pittidae have a rounded body and a short bill, and many species are brilliantly plumaged. Both adults incubate the eggs.



20 cm
8 in

BLUE-WINGED PITTA

Pitta moluccensis
Found from S. China to Borneo and Sumatra, the blue-winged pitta inhabits dense forest when breeding, but overwinters in coastal scrub.



19 cm
7½ in

INDIAN PITTA

Pitta brachyura
Like its relatives, this pitta from the S. Himalayas, India, and Sri Lanka builds domed nests on or near the ground.